# Session Objectives

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

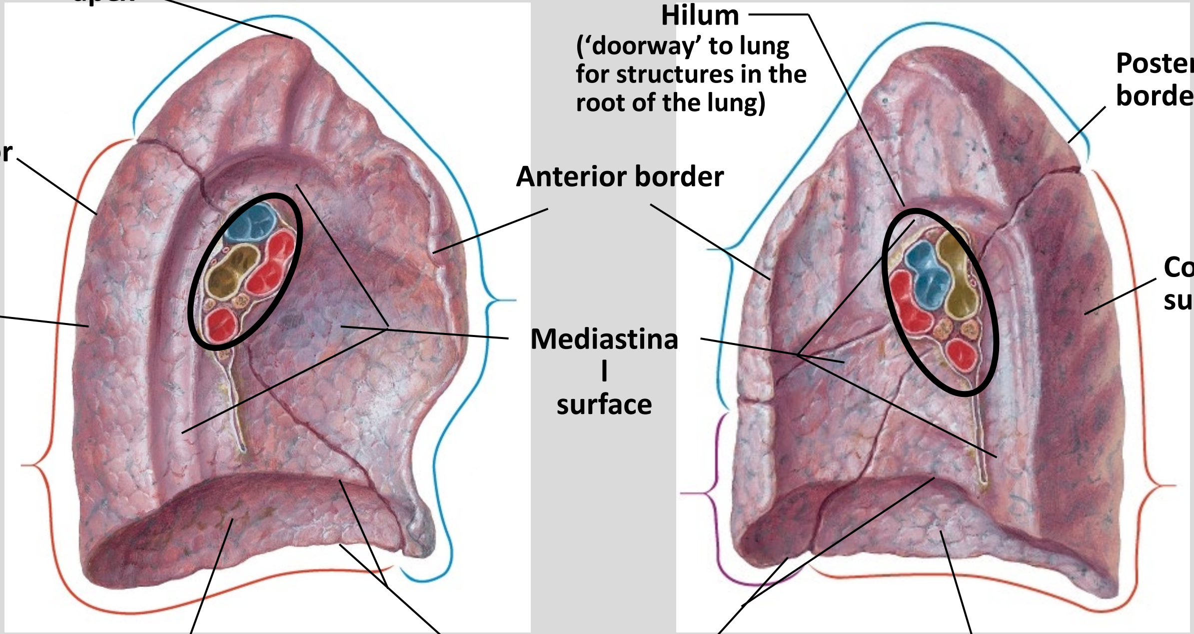
1. Describe the gross anatomy of the respiratory system
2. Explain the embryological origins of the parts of the respiratory anatomy
3. Recognize the tissue types found in the respiratory system and understand its embryological origins.
4. Understand how the development of the lung leads to its common functional and clinical circumstances

Source: Course Syllabus

Surfaces:
- mediastinal
- diaphragmatic
- costal

Borders:
- anterior
- posterior (smooth)
- inferior

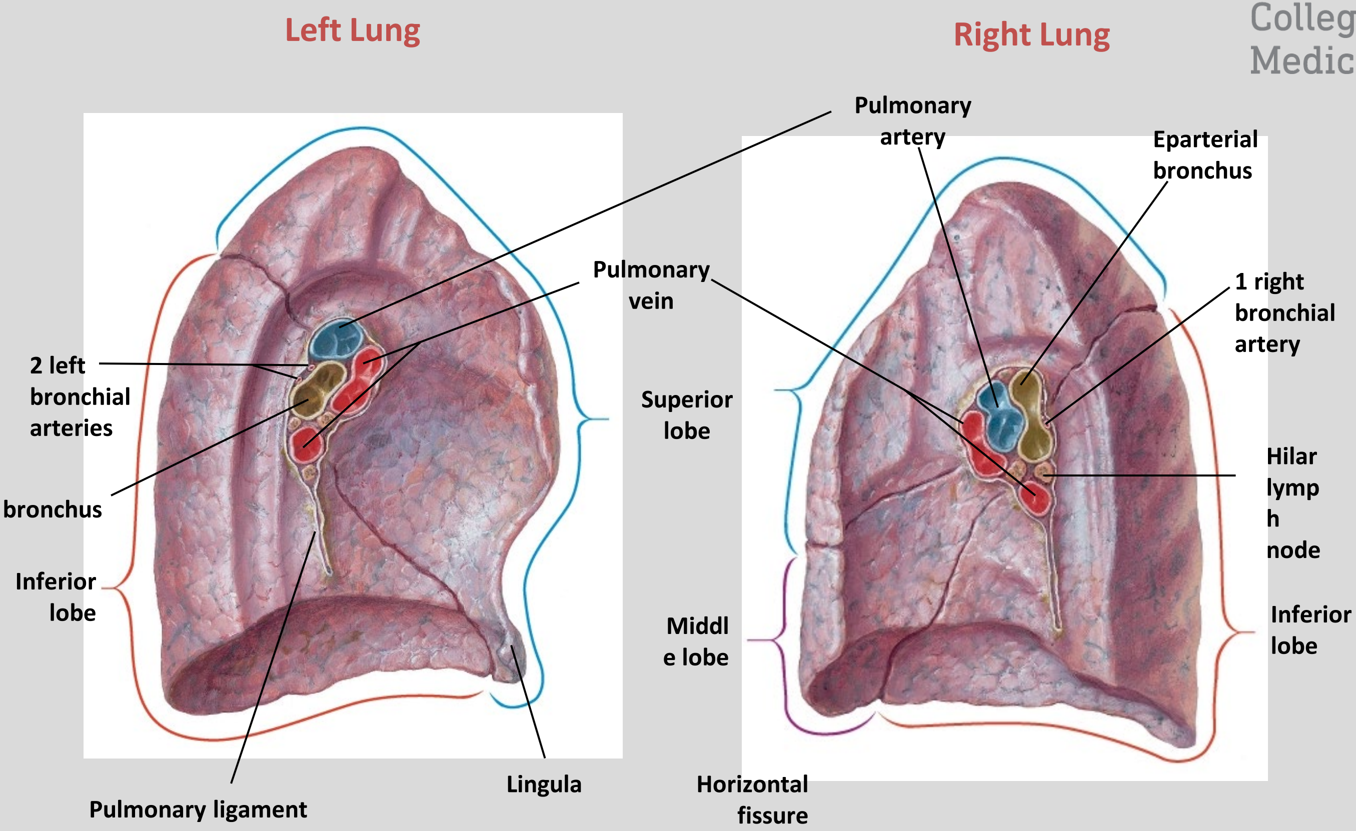




# THE ROOT OF THE LUNGS

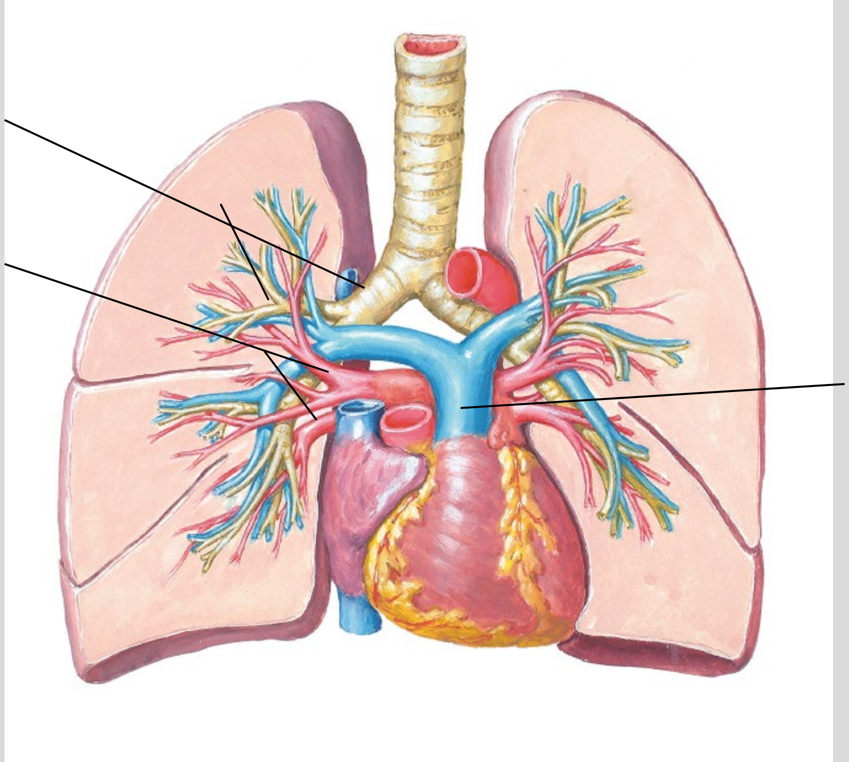
NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine



NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

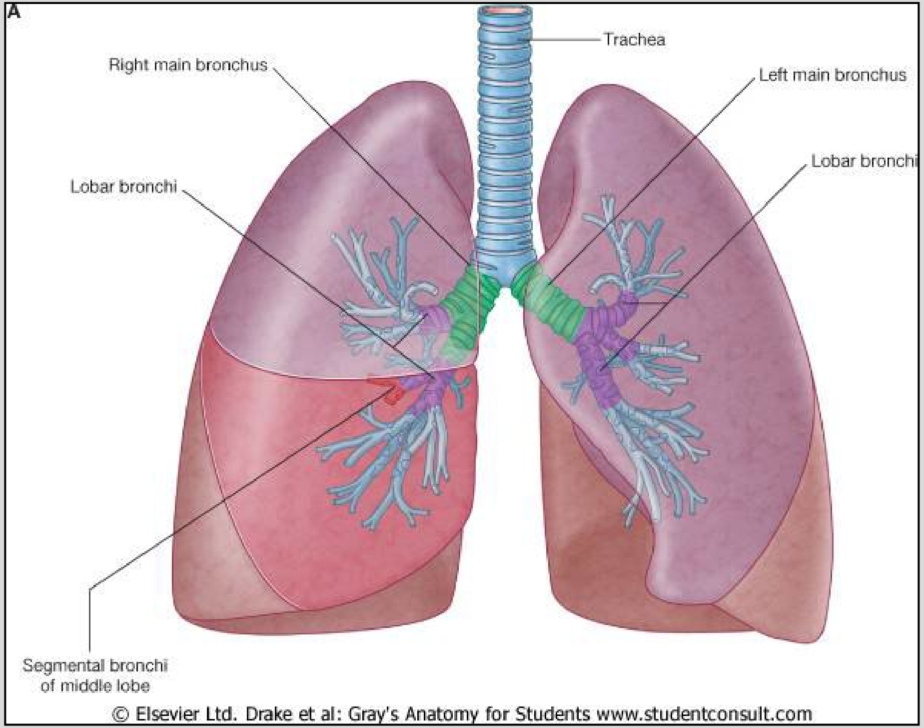



- Pulmonary arteries (one per side) run alongside the bronchial tree into each bronchopulmonary segment (intrasegmental)
- Pulmonary veins (two per side) are intersegmental and drain newly oxygenated blood from adjacent bronchopulmonary segments

# Trachea and lungs

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine



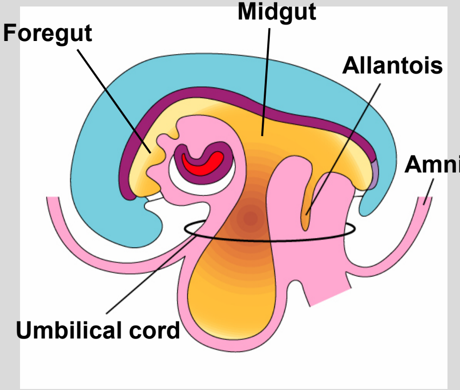

- The trachea is a tube composed of C-shaped cartilaginous rings, closed off posteriorly by smooth muscle

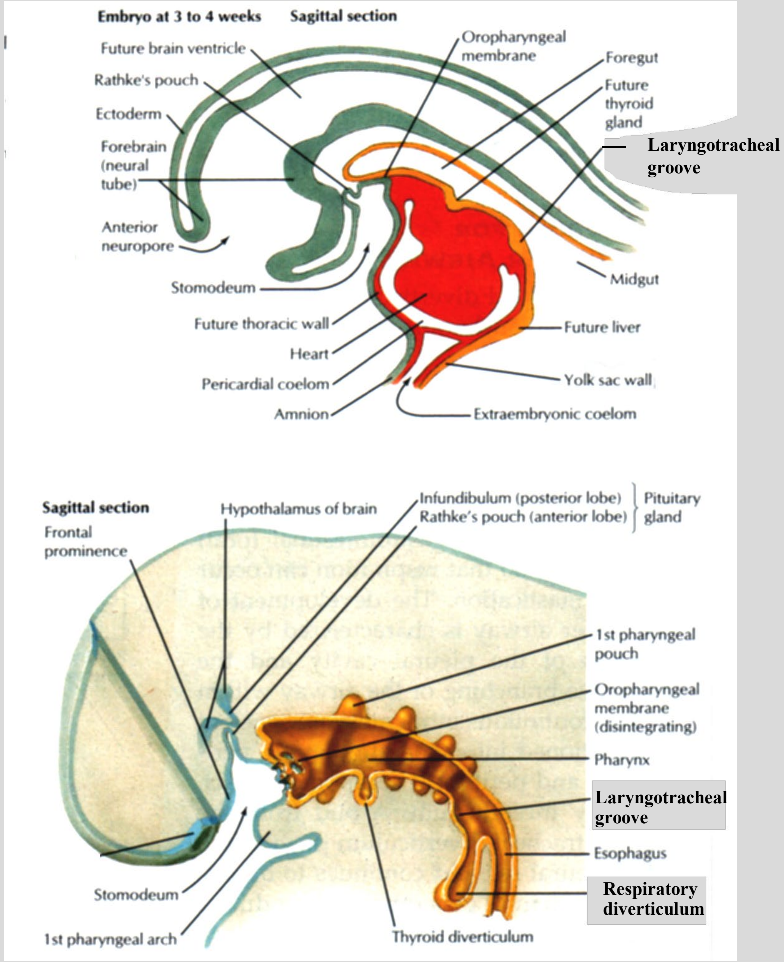
-Runs from C6 to T4/T5, where it bifurcates into left and right mainstem (primary) bronchi. The right mainstem bronchus is shorter, wider and more vertically oriented than the left bronchus. Therefore, aspirated objects are more likely to lodge on the right side.

-Primary bronchi enter the lung after which they further divide into secondary (lobar) bronchi (3 right, 2 left)

-Lastly, secondary bronchi divide into tertiary (segmental) bronchi (10 right (3 upper, 2 middle, 5 lower), 10 left (5 upper, 5 lower))

# LARYNGOTRACHEAL GROOVE & RESPIRATORY DIVERTICULUM




NEW YORK INSTITUTE OF TECHNOLOGY

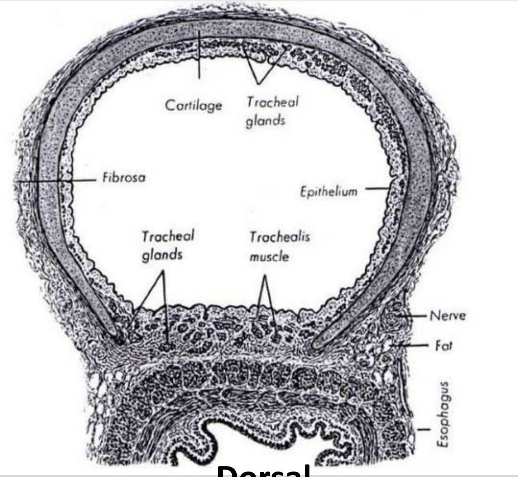
College of Osteopathic Medicine

lungs begin to develop in the 4th week from the foregut splanchnopleure in the region of the primordial pharynx.

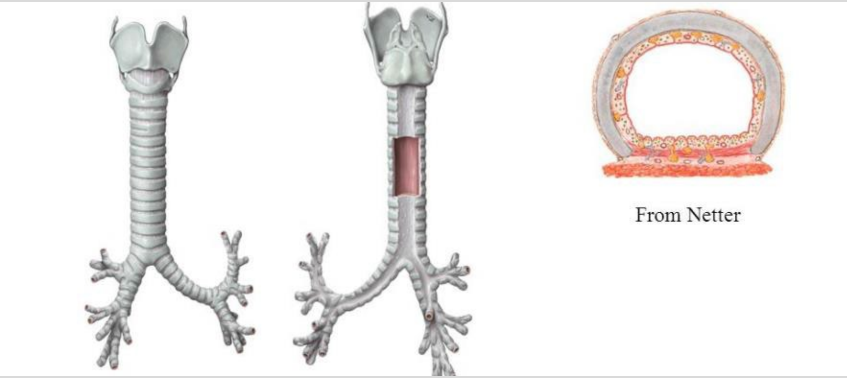
-**Laryngotracheal groove -** median outgrowth from caudal end of ventral wall of primordial pharynx (foregut). It is the primordium of the lower respiratory tract.

-By the fifth week, the laryngotracheal groove expands into a pouchlike **respiratory diverticulum.** This diverticulum grows caudally into the splanchnic mesoderm, paralleling the esophagus.

Ventral




Dorsal


From Netter

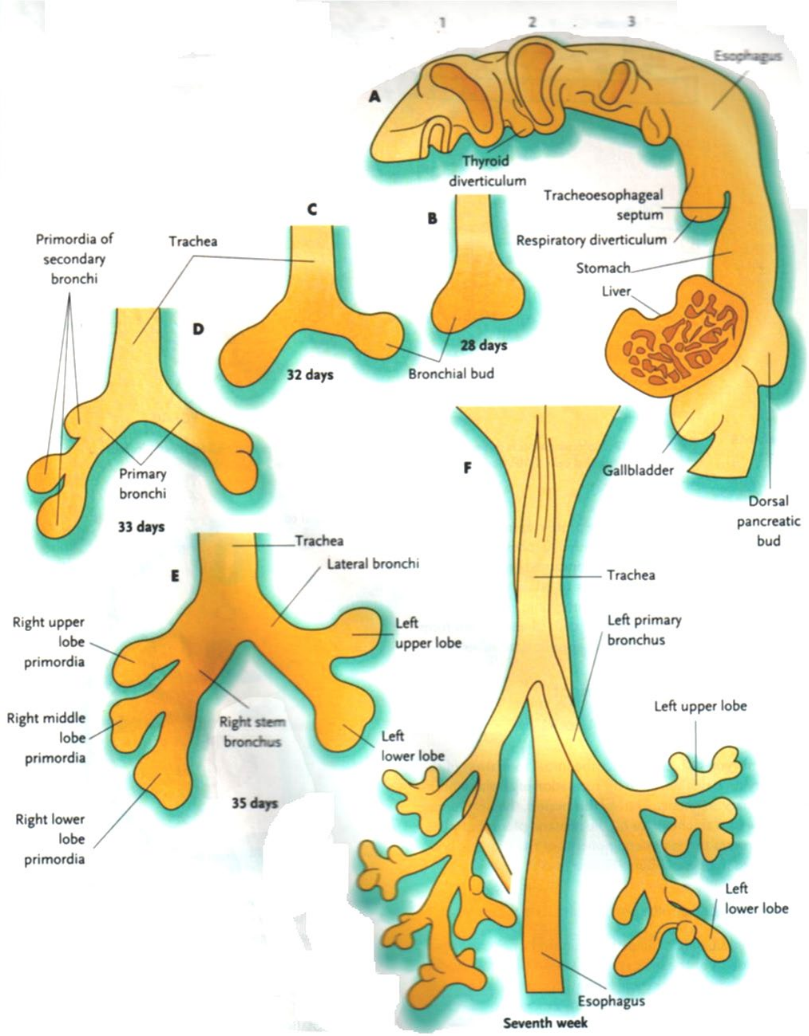
# Trachea

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

**Structure of the airways varies considerably Depends on their location in the tracheobronchial tree**

**TRACHEA**

- **Fibromuscular tube**
- **Supported by c-shaped cartilage & by smooth muscle.**
- **Multiple cartilage rings extend about 5/6ths of the way around the trachea**




# BRANCHING OF THE BRONCHIAL BUDS

**NEW YORK INSTITUTE OF TECHNOLOGY**

College of Osteopathic Medicine

-The distal end of the respiratory diverticulum expands and soon divides into into two **lung/ bronchial buds.** These buds will become the bronchial tree and lungs. The part of the diverticulum proximal to the first bifurcation becomes the **trachea and larynx.**

-This first branching (into two bronchial buds) is for the **main stem/ primary bronchi**.

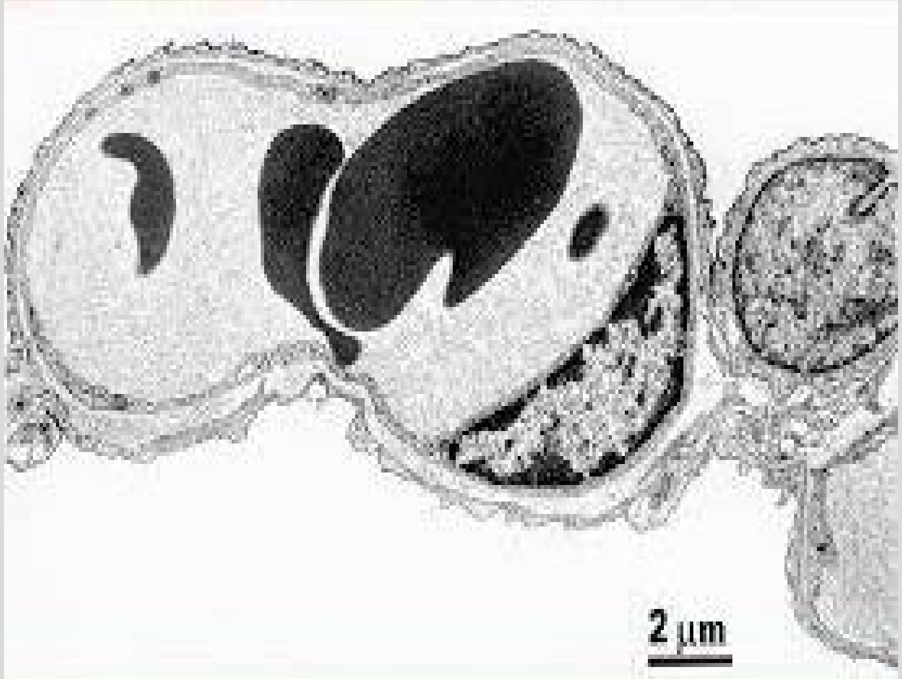
-The second branching is for the **lobar/ secondary bronchi** (2 left, 3 right) which give rise to the lobes of each lung.

-The third branching takes place in the 6$^{th}$ week and yields 10 **segmental/ tertiary bronchi** per lung. These segmental bronchi will give rise to **bronchopulmonary segments** in the adult lung.

-Branching continues throughout the fetal period.

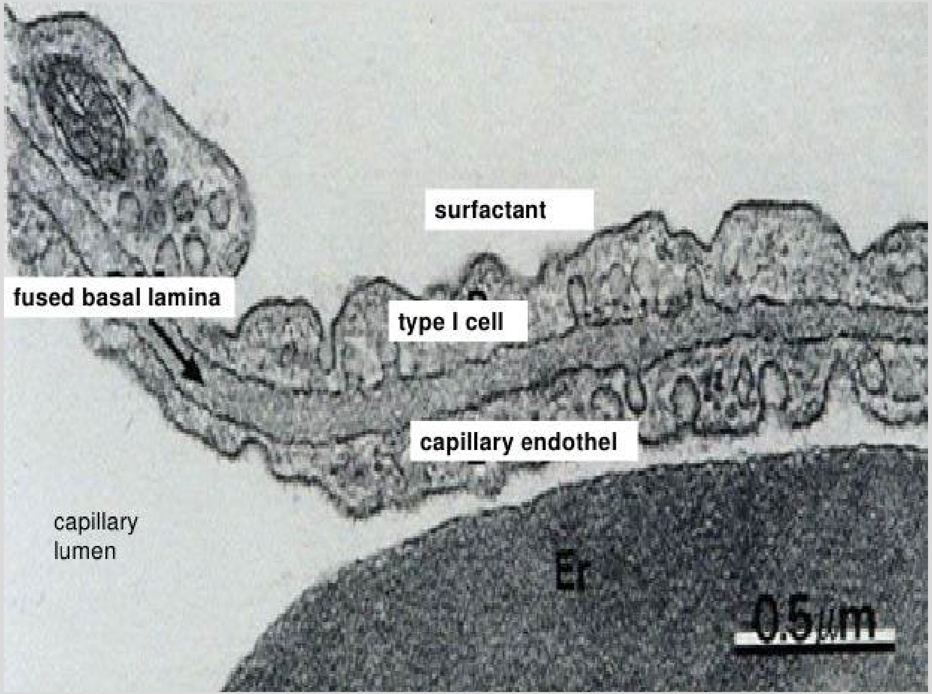
**Endoderm:** epithelia and glands of larynx, trachea, bronchi, pulmonary epithelium

**Splanchnic mesoderm:** pulmonary connective tissue, cartilage and smooth muscle




NEW YORK INSTITUTE OF TECHNOLOGY

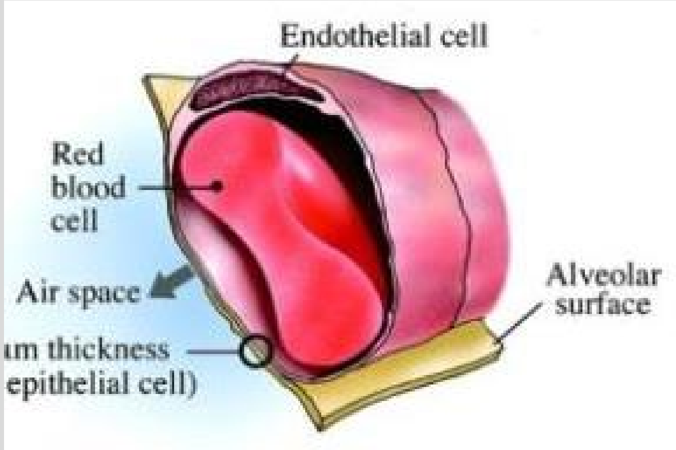
College of Osteopathic Medicine



# Pulmonary Capillaries

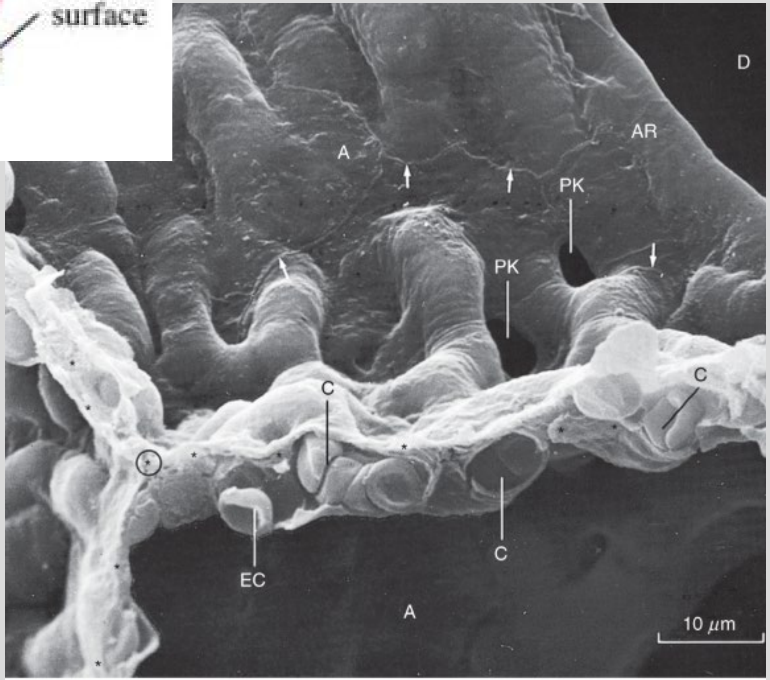
NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine




**<u>PULMONARY CAPILLARIES</u>**

- Capillary wall is extremely thin
- ~ 250 to 300 ml/m$^2$ of blood the pulmonary circulation
- 60 to 70 m/m$^2$ located in the pulmonary capillaries
- Serve as a blood reservoir
- RBC: ~ 4 to 5 s to travel through the pulmonary circulation (at rest)
- ~ 0.75 s in pulmonary capillaries
- Average diameter: ~6 µm
- Average RBC: ~8 µm
- RBC change shape as they pass through the pulmonary capillaries
- 280 billion pulmonary capillaries supply ~ 300 million alveoli

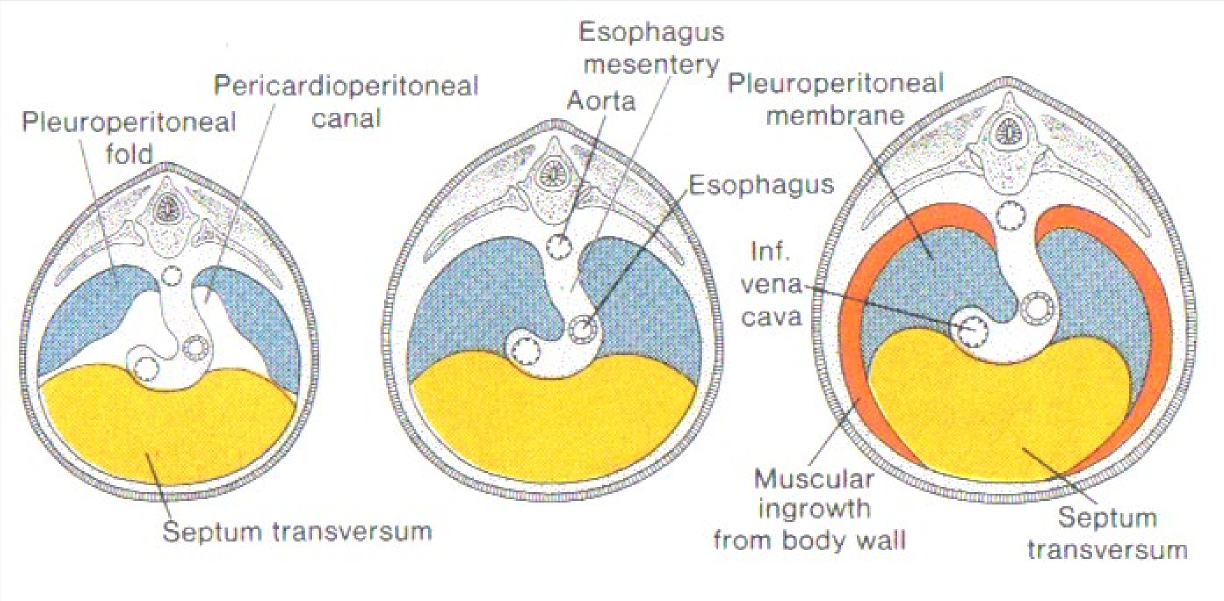



Source: Levitzky MG: *Pulmonary Physiology, Eighth Edition*: www.accessmedicine.com

# DEVELOPMENT OF THE DIAPHRAGM

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine




-As the pleuroperitoneal folds expand, myoblasts from the body wall migrate in. Eventually the folds fuse with the dorsal mesentery of the esophagus and the septum transversum. Together, all of these structures contribute to the formation of the diaphragm, and separate the pleural and pericardial cavities from the peritoneal cavity.

- -Septum transversum (originates cranial to the heart in the trilaminar embryo)
- -Pleuroperitoneal membranes
- -Dorsal mesentery of the esophagus
- -Muscular ingrowth from lateral body walls

# PLEURAL CAVITIES



- The developing lungs grow into the portion of the pericardioperitoneal canals that is to become the pleural cavity.
- The layer of coelomic epithelium that invests the tissue of the lungs is called the visceral pleura. The layer that lines the walls of the thoracic cage is called the parietal pleura, The space enclosed between the visceral and parietal pleurae is the pleural cavity– it contains nothing but a thin layer of serous fluid which allows for free expansion and contraction of the lungs.  The two layers of pleura are continuous at the hilum of the lung.
- Parietal pleura
    - separated from the thoracic wall and diaphragm by the endothoracic fascia.
    - named according to the structures that it abuts (costal, diaphragmatic, mediastinal, cervical)
- Costodiaphragmatic recesses
    - Regions where inferior aspects of costal pleura are in contact with the diaphragmatic pleura without the presence of any intervening lung tissue.
    - During pathologies, substances may collect in the costodiaphragmatic recesses (e.g.,.pneumothorax, haemothorax, hydrothorax) and a pleural tap/ thoracocentesis may be necessary. Such a procedure should be performed in the $8^{th}$ or $9^{th}$ IC space just superior to the superior border of the rib.
    - occur anteriorly where costal pleura is opposed to mediastinal pleura

# Lungs

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Both lungs:

- apex
- mediastinal, diaphragmatic and costal surfaces
- anterior, inferior and posterior borders
- 1 primary bronchus (the right is wider, shorter, more vertical); 10 tertiary bronchi (bronchopulmonary segments)
- **root of the lung through the hilus**. This includes these structures:
    - pulmonary artery (intrasegmental) -
    - primary bronchus (intrasegmental)
    - pulmonary veins (intERsegmental)
    - Lymphatics
    - bronchial arteries (2 left, 1 right)
    - pulmonary plexus of nerves (ANS)
    - bronchial veins (to azygos and hemiazygos veins)

The left lung:

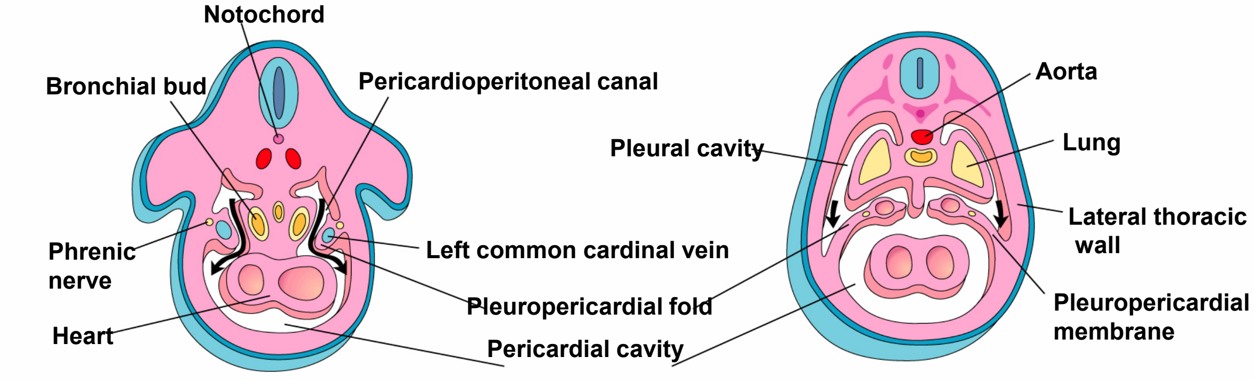
- two lobes/ secondary bronchi (superior and inferior)
- one fissure (oblique)
- large cardiac impression and notch
- lingula – process of upper lobe just inferior to the cardiac notch

The right lung:

- three lobes/ secondary bronchi (superior, middle and inferior)
- two fissures (horizontal and oblique)

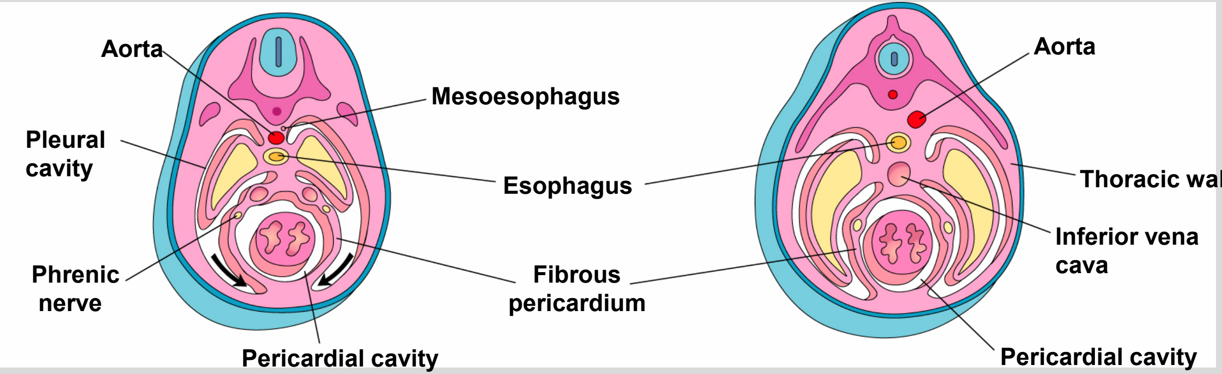
# PLEUROPERICARDIAL FOLDS/MEMBRANES

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine



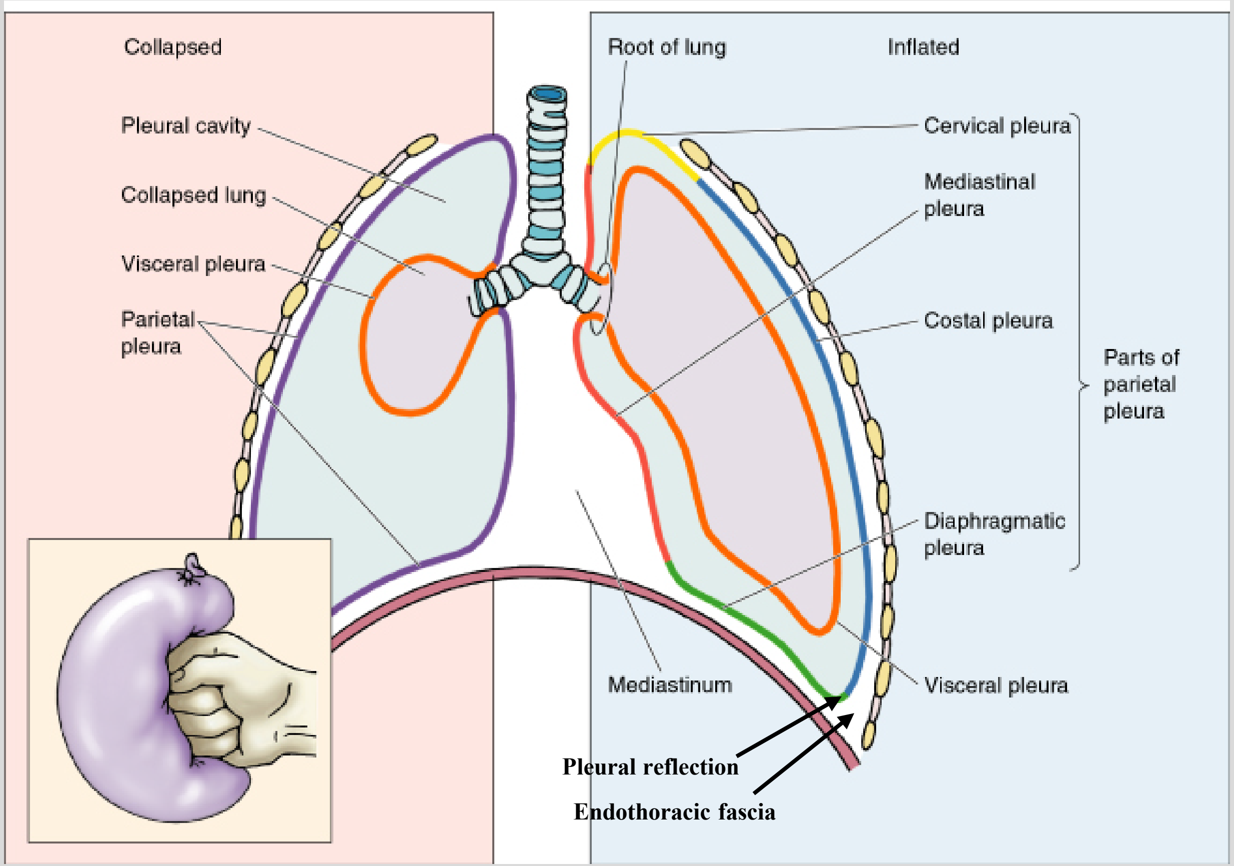

Week 5

Week 6




Week 7

Week 8




# NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

-the lungs develop by invaginating into a closed pleural cavity. This results in lungs surrounded by a collapsed sac that contains only a small amount of lubricating serous fluid.

- The layer of the pleural sac that immediately invests the lung tissue is called the visceral pleura.
- The outer layer is called the parietal pleura (costal, mediastinal, diaphragmatic, cupola/cervical).